

Astrolabe

Calibrated reconstruction of a Parkinson’s motor diary
from bilateral wrist accelerometry

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Abstract

Parkinson’s symptoms fluctuate hour to hour, but the paper diary intended to capture that is abandoned within days: in the cohort studied here only 61.6% of possible hours carry an entry, *with* research staff supporting the process. We ask whether wrist accelerometry can fill the gaps and, more importantly, whether a system can know when it cannot.

We report two results of opposite sign. Tremor presence is recoverable across people (held-out AUC = 0.697 ± 0.075). The seven-point subjective motor scale is not: nine feature sets over five participant-level folds all fail to beat a constant predictor, and the failure mode is a sign flip in the feature–state relationship between individuals rather than insufficient capacity. We therefore do not predict it.

The contribution is the uncertainty machinery around both. Credible intervals are widened on held-out participants until empirical coverage matches the nominal level (0.903 against 0.90), abstention is tuned to a fixed *error budget* rather than a fixed rate, and selective prediction is measured rather than asserted: accuracy rises from 0.713 to 0.825 as the answered fraction falls to a quarter. Removing one wrist forces abstention from 12.4% to 77.3% to hold the same budget — degradation the system discovers rather than is told.

1 Data

COPS (Continuous Observation of Parkinsonian Symptoms; Nesser et al., CC-BY 4.0) provides 66 participants wearing GENEActiv accelerometers on both wrists at 100 Hz, paired with hourly self-reported symptom diaries over roughly a week each. After excluding participants without usable paired data, 6,530 hours across 65 participants remain.

The alignment convention. A diary row stamped T describes the window $[T - 1, T)$. This is stated once in the study documentation and is easy to invert; doing so shifts every label by one hour and silently degrades every downstream result. `io_cops.py` re-derives the window from the filename and raises on mismatch, so a misaligned join cannot be constructed.

Cohort caveat. 46 of 66 participants have deep brain stimulation. This is an advanced, device-treated population; both the movement signature and the fluctuation pattern differ from newly diagnosed patients, and nothing here should be assumed to transfer unchanged.

2 Method

2.1 Ordinal emissions

The kinesia scale is ordinal with $K = 7$ levels. Following Frank and Hall we decompose it into $K - 1$ cumulative binary problems, fit a gradient-boosted classifier to each, and difference the results:

$$\Pr(z = k \mid x) = \Pr(z > k - 1 \mid x) - \Pr(z > k \mid x), \quad (1)$$

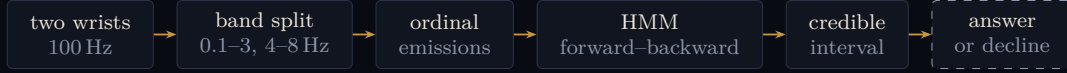


Figure 1: The inference chain. The final stage is a decision, not a prediction: where the interval is too wide to stand behind, nothing is emitted.

enforcing monotonicity of the cumulative estimates by a running minimum before differencing, so no negative mass can appear.

2.2 Dividing out the prior

A discriminative classifier returns $p(z \mid x)$, but the state-space model requires an emission likelihood $p(x \mid z)$. By Bayes' rule, up to a factor constant in z ,

$$p(x \mid z) \propto \frac{p(z \mid x)}{p(z)}. \quad (2)$$

Omitting the division counts the class prior twice — once in the classifier and once in the chain — biasing the posterior toward the majority state in a way that resembles smoothing working well.

2.3 Temporal model

Motor state is persistent, so we place a hidden Markov model over the day and recover $\gamma_t(z) = \Pr(z_t = z \mid x_{1:T})$ by forward-backward in log space. The transition matrix is a convex combination rather than a matrix root,

$$T = (1 - s)I + sB, \quad s = 0.02, \quad (3)$$

with B the empirical hour-to-hour transition matrix. A matrix root of B can contain negative entries and is not a valid stochastic matrix at fine steps. Likelihoods are tempered before smoothing, $\tilde{\ell} = \tau^{-1} \log p(x \mid z)$ with $\tau = 25$ selected on a held-out fold; without it the day collapses onto a single state.

2.4 Calibrated credible intervals

For posterior γ we take the shortest contiguous interval carrying mass m :

$$[\ell, u] = \arg \min_{u-\ell} \left\{ \sum_{k=\ell}^u \gamma_k \geq m \right\}. \quad (4)$$

Crucially m is *not* set to the nominal 0.90. Requesting 90% of posterior mass does not deliver 90% empirical coverage, and asserting otherwise is exactly the unfalsifiable claim this system exists to avoid. Instead m is swept on participants held out of *both* the emission fit and the test fold until achieved coverage reaches target:

$$m^* = \min \left\{ m : \frac{1}{n} \sum_i \mathbf{1}[z_i \in [\ell_i(m), u_i(m)]] \geq 0.90 \right\}, \quad (5)$$

giving $m^* = 0.72$ and achieved coverage 0.903.

A bug worth reporting. Interval mass was initially fitted on the raw emission posterior and applied to the smoothed HMM output. The two distributions have entirely different sharpness, and coverage landed at 0.535 against a 0.90 target — which the interface would have displayed as “90%”. Calibration and inference now share one code path so they cannot diverge.

2.5 Abstention as a fixed error budget

A step is declined when the posterior is uninformative:

$$\text{abstain} \iff \max_k \gamma_k < \pi \quad \vee \quad (u - \ell + 1) \geq w. \quad (6)$$

The threshold π is *not* tuned to a target abstention rate. Doing so guarantees every sensor configuration declines equally often, so a degraded configuration cannot exhibit degradation. We observed exactly this: with one wrist blanked, abstention moved the wrong way, because gradient-boosted trees route missing features down a default branch and return a *sharper* posterior on strictly less evidence.

Instead π holds a fixed risk on answered steps. With $\mathcal{A}(\pi)$ the answered set and $\hat{z}_i = \gamma_i^\top(0, \dots, 6)$,

$$\pi^* = \min \left\{ \pi : \frac{1}{|\mathcal{A}(\pi)|} \sum_{i \in \mathcal{A}(\pi)} |\hat{z}_i - z_i| \leq \beta \right\}. \quad (7)$$

A worse configuration can only meet β by answering less, so the abstention rate becomes a consequence of measured accuracy.

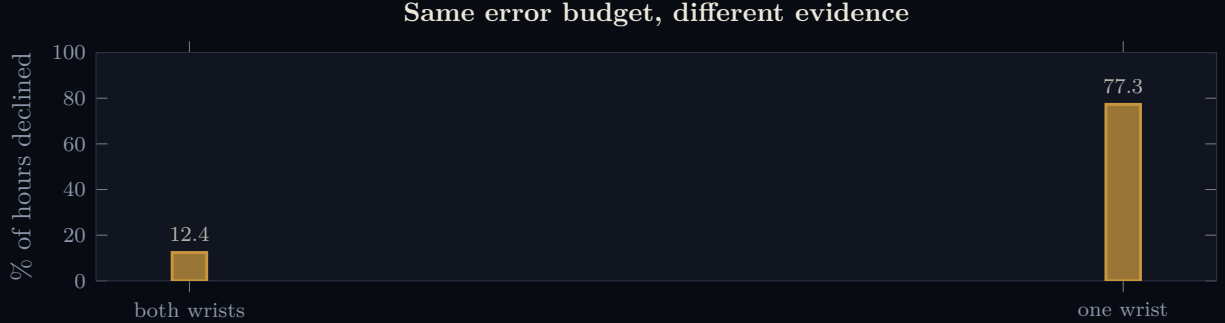


Figure 2: Holding MAE on answered steps fixed at $\beta \approx 0.22$, removing one wrist forces the system to decline six times as many hours. Measured on 6,838 hours from 11 held-out participants.

3 Results

Target	Metric	Held-out
Tremor present (TremorScore > 0)	AUC	0.697 ± 0.075
Kinesia OFF/ON, identical folds	AUC	0.567
Kinesia, 7-state reconstruction	ordinal MAE	0.684
constant baseline (always “Good”)	ordinal MAE	0.594
per-participant oracle	ordinal MAE	0.380

Table 1: Tremor is a 4–8 Hz mechanical oscillation the accelerometer measures directly. How well an hour *felt* is a judgement the sensor has no access to.

3.1 Why the subjective scale fails

The cumulative classifiers reveal the mechanism. Held-out AUCs for $\Pr(z > k)$ are 0.428, 0.530, 0.587, 0.324, 0.455 — three of five *below chance*. A classifier at 0.5 has no signal; one at 0.324 has signal pointing the wrong way. The feature-to-state mapping inverts between individuals, and no amount of regularisation repairs a sign that flips.

Personalisation does not rescue it. Given identical hours and the *same* calibration labels, a participant’s own constant scores ordinal MAE 0.400 while the global model plus a per-participant offset scores 0.474. The offset was learning the person’s mean; the global mapping it shifted added nothing.

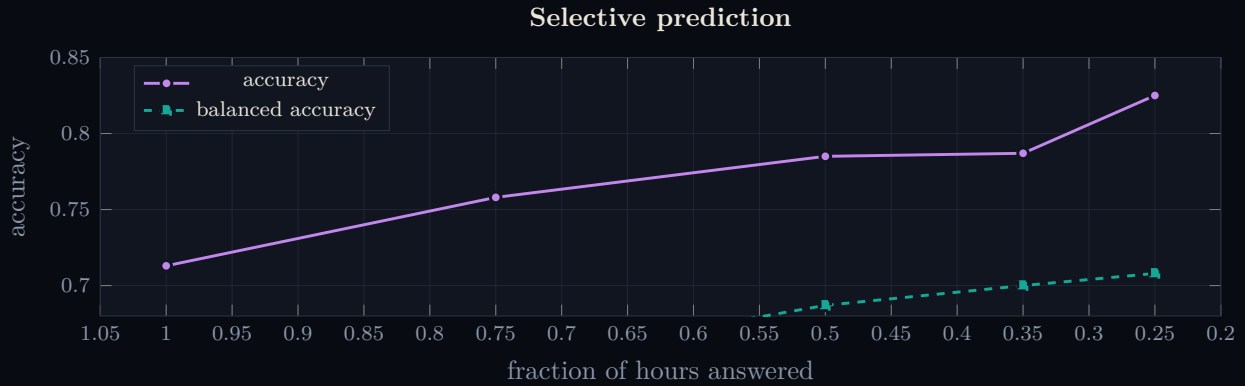


Figure 3: Accuracy rises as the answered fraction falls: the hours the model declines really are the ones it would have got wrong. Balanced accuracy is shown because at 30.5% prevalence raw accuracy can be bought by predicting the majority class.

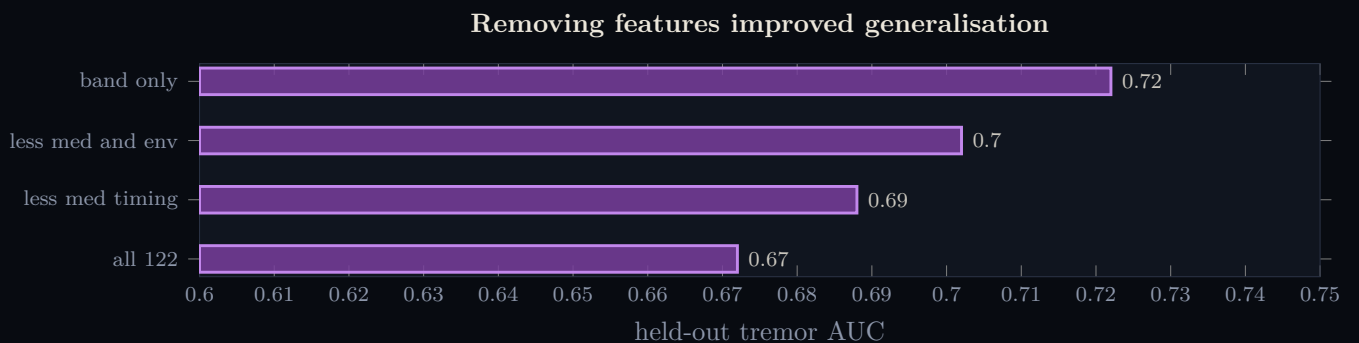


Figure 4: Medication-timing features alone score $AUC = 0.433$, below chance: who takes which drug on what schedule is close to a participant fingerprint. The shipped detector reserves a fifth of training participants for calibration, costing ≈ 0.025 AUC and giving the 0.697 quoted throughout.

3.2 The caveat that must be stated

Pooled out-of-fold AUC is 0.681, but the median *within-participant* AUC is 0.550, above chance on 25 of 36 scorable participants. Much of the pooled figure separates tremor-dominant people from everyone else rather than a tremulous hour from a calm one *in the same person* — and within-person discrimination is what a diary requires. Both figures ship in the output contract.

4 What the system will not claim

The seven-point motor scale is never inferred. It is displayed as reported by the patient, or as an explicit abstention. On the demonstration participant-day the model declines all 114 steps, its peak posterior never exceeding 0.549 against a threshold of 0.58 fixed on other participants. That is the system behaving correctly.

No causal claim is made. Trajectory shape around recorded medication times is temporal alignment, not treatment response; establishing response requires a controlled protocol. This is not a medical device.

5 Where this goes next

The negative result is the most informative thing we have, and it points at specific work rather than at more capacity.

Per-person recalibration from a handful of labels. The detector is calibrated to a cohort prevalence of 0.30; a participant at 0.74 receives probabilities systematically below their own base rate, so ranking is preserved while the operating point is wrong. Asking a user to label five to ten hours once would let an isotonic fit be re-estimated per person. This is the cheapest available improvement and it converts the within-person weakness from a caveat into a workflow.

Hierarchical priors instead of one global model. The sign flip in $\Pr(z > k)$ between individuals is exactly the structure a partially-pooled model addresses: person-level coefficients drawn from a cohort prior, so each participant borrows strength without being forced onto a shared mapping that does not exist for them. A hierarchical ordinal model is the principled version of the per-participant offset that failed here.

Conformal prediction in place of the coverage sweep. Our interval mass is fitted empirically, which yields marginal coverage but carries no finite-sample guarantee. Split conformal over participant-level folds would give distribution-free validity, and *Mondrian* conformal — conditioning on sensor configuration — would deliver the per-configuration guarantee we currently obtain by refitting.

Learning the abstention rule rather than thresholding it. Selective prediction is currently a threshold on peak posterior. Training a selector jointly with the predictor, against a cost that prices a wrong answer above a refusal, would let the model learn *which kinds* of hour to decline rather than only how confident to be.

Targets the sensor can actually reach. Dyskinesia has a distinct spectral signature and is plausibly recoverable by the same machinery. Medication-state transitions — the onset of an ON period — are a change-point problem rather than a classification one, and matter more clinically than an absolute state. Both are better uses of this signal than another attempt at the subjective scale.

Prospective validation. Everything here is retrospective on one cohort of advanced, largely DBS-treated patients. The claim that matters — that a diary which refuses is more useful than one that guesses — is a claim about clinical workflow, and only a prospective study with clinicians using it can test it.

6 Reproducibility

Script	Establishes
<code>baselines.py</code>	the constant and oracle baselines
<code>diary_completion.py</code>	the 61.6% completion rate
<code>diagnose_deep.py</code>	tremor versus kinesia on identical folds
<code>diagnose_featureset.py</code>	the nine-feature-set sweep
<code>diagnose_fair.py</code>	offset versus constant given equal labels
<code>train_tremor.py</code>	the shipped detector and selective curve
<code>make_bundles.py</code>	calibration, abstention, and the output contract

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aggedright extbfData. Nesser, T. et al., *COPS — Continuous Observation of Parkinsonian Symptoms*, OSF (2025). doi:10.17605/OSF.IO/5XVWN, CC-BY 4.0.

extbfCode. MIT. github.com/MohammadSadeghSalehi/astrolabe

extbfStatus. Not a medical device. No diagnostic, dosing or treatment claim is made.